Assignment 2

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Section A

Q1

(i) Network A supports 1022 hosts

For 1022 hosts:

$$2^{10}-2 = 1022.$$

so, Network A uses a /22 subnet mask

A /22 subnet increases by 4 in the 3rd octet.

Given source IP:

172.16.8.1.

Network A is: 172.16.8.0/22.

Range :

Network address : 172.16.8.0

Broadcast address : 172.16.11.255

since Network B is the next consecutive subnet :

$\boxed{172.16.12.0/22}$

The destination ID is :

172.16.13.255

which belongs to subnet 172.16.12.0/22

Hence, Network B's subnet address is :

$\boxed{172.16.12.0/22}$

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ii) Maximum number of hosts Network B can support

Network B is 9180/22.

Host Bits: $32 - 22 = 10$.

maximum hosts: $2^{10} - 2 = 1022$

$\boxed{1022 \text{ hosts}}$

iii) VLSM subnetting.

IP Block : 10.192.128.0/18.

- subnet 1 (1048 hosts) : Needs. 11 bits

$2^{11} = 2048$. mask : /21.

Address : 10.192.128.0/21

(Range : 128.0 –135.255)

- subnet 2 (250 hosts) : Needs 8 bits

($2^8 = 256$) . Mask : /24.

. Address : 10.192.136.0/24

(Range : 136.0 – 136.255)

- subnet 3 (62 hosts) : Needs 6 bits

($2^6 = 64$) Mask : /26

Address : 10.192.137.0/26

Range : (137.0 – .137.63)

<u>Q2</u>

Given,

original datagram size = 4680 byte

Header size = 20 bytes.

Data size:

$4680 - 20 = 4660$ bytes.

Different between fragment offset : 75.

fragment offset unit = 8 bytes.

Therefore, data carried by each fragment:

$75 * 8 = 600$ bytes

i.) calculate MTU.

- Data = 600 bytes
- Header = 20 bytes.

Therefore, MTU = 600+20

=> 620 bytes.

ii) fragment offset of the 4th fragment

offset :

1st fragment = 0

2nd fragment = 75.

3rd fragment = 150

4th fragment = 225.

iii) Data size of the last fragment

Total data: 4660 bytes.

Each full fragment carries:

600 bytes

Number of full fragments:

~~4600 ÷~~

$4660 \div 600 = 7$ full fragment with remainder.

Data sent by 7 fragments:

$7 * 600 = 4200$

Remaining: $4660 - 4200 = 460$.

Therefore last fragment data size: $\boxed{\text{460 bytes}}$

iv) If DF bit was set.

DF = Don't fragment.

If DF is set and packet size exceeds MTU:

→ Router drops the packet.
→ Router sends ICMP "fragmentation Needed" message back to sender
→ sender reduces packet size using path MTU Discovery.

packet would be dropped and ICMP fragmentation needed would be sent.

Q3

i) The first DHCP message is DHCP Discover.

Source IP: $\boxed{0.0.0.0}$

Destination IP: 255.255.255.255

ii) Although both DHCP Discover messages are broadcast, each host has a unique

- MAC address.
- Transaction Id (xid)

DHCP server uses these field to identity clients separately.

DHCP Distinguishes them using unique mac address and transaction IDs.

iii) if DHCP-2 doesn't respond during renewal:

1. Less renewal fails.
2. client enters rebinding state.
3. Eve broadcasts DHCP Discover/ Request.
4. Any available DHCP server may respond.
5. DHCP-1 offers a new address.

~~For~~

Eve broadcast a new DHCP request & DHCP-1 responds with a new lease

iv) Before sending to Router R_1 Asper must know Router R's MAC address.

1. Asper sends ARP request

2. Router R replies with a ARP reply containing its MAC address.

3. Asper encapsulates packet using:

- Destination MAC = Router R MAC
- source MAC = Asper MAC

ARP request & ARP reply are exchanged

Section B

Q5

i) A- non - recursive static route uses the ~~exit~~ exit interface.

General syntax:

ip route <destination-network> <mask> <exit- interface>

Example:

ip route 192.168.2.0.255.255.0 s0/0/0.

ii) A floating route has a higher administrative distance.

Example:

ip route 0.0.0.0 0.0.0.0 S0/0/0 200

Here,
200 = administrative distance
Route is used only if primary route fails.

Q5

i) A. FE80::1A2B.

Expanded:

FE80:0000:0000:0000:0000:0000:0000:1A2B.

B. 2001: DB8: 1234::10

Expanded:

2001: 0DB8 : 1234 : 0000 : 0000 : 0000 : 0000 : 00..

C. ::25

Expanded

0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0025

<u>ii)</u> 224.10.10.5 is a multicast IP address. only multicast group members receive multicast packets.

Q6

i) Asif sending frame to Eve

since Asif doesn't know Eve's MAC:
He sends an ARP request.

Frame:
. source MAC = Asif
. Destination MAC = FF:FF:FF:FF:FF:FF (broadcast)

switch behavior:
. S1 learns Asif MAC on incoming port
. Floods broadcast frame.
. S2, S3, S4 also flood broadcast frame

· Eve receives ARP request.

ii) MAC tables after Eve replies.

→ switches ~~can~~ learn Eve MAC on corresponding ports.

→ switches already learned Arif MAC earlier.

Therefore switches store:

switch	learned MACs
S1 →	Arif, Eve
S2 →	Arif, Eve
S3 →	Arif Eve.
S4 →	Arif, Eve.

Section C

i) **Q7** TTL prevents packets from circulating forever due to ~~route~~ routing loops.

Each router decrements TTL by 1.

When TTL ~~ree~~ reaches 0:

- Router discards the packet.
- Sends ICMP Time exceeded.

without TTL:

- Routing loops would create infinite traffic.
- Network congestion would grow uncontrollably.
- Internet performance would collapse

ii) This is smurf Attack.

Attackers

1. sends ICMP Echo request to broadcast address

2. spoofs victim's IP as source

3. All active hosts reply to victim

broadcast amplification with spoofed victim IP causes 200 replies.

Q8

i) No.

reason :

1. link state protocols do not suffer from count to infinity

2. Routes using LS immediately flood ~~top~~ topology changes.

Distance vector routers may suffer from it. but LS routers prevent propagation.

ii) when link fails:

→ Routers remove learned routes

→ new route calculations have not completed ~~out~~ yet

→ until convergence finishes only directly connected routes remain.

Q 2

i) PAT uses:

same public IP.

Different source ports.

220. 20. 20. 17/30

public IP used:

220. 20. 20. 17.

Host A becomes:

. source IP = 220. 20. 20. 17.

. source port = translated unique port.

Host D becomes:

. Source IP = 220.20.20.17.

. source port = another unique port

Host	translated source IP	translated port
A	220.20.20.17	unique port
B	220.20.20.17	different unique port.

ii) No. PAT allows outbound session only.
External host ~~connec~~ cannot directly reach internal private hosts because:
→ private IPs are not routable.
→ No NAT entry exists.

To make it possible:

→ configure static NAT or port forwarding

Q10

1.) IPV6 transition technique.

scenario 1 requires IPV6 to IPV4 communication.

$\boxed{\text{Dual stack}}$

ii) IPV6 addressing mechanism in scenario 2

using same IPV6 address on multiple routers. $\boxed{\text{Anycast Addressing}}$

Difference from multicast

. Anycast → delivered to nearest on device

. Multicast ⇒ delivered to all group members.

(ii) IPV6 uses:

$\boxed{\text{Extension Headers}}$

Examples:

→ Hop by hop options
→ Routing Header
→ fragment Header
→ Destination options.

. These headers are chained after the fixed 40 byte base header.

Q11

i) check the least significant bit of the octet

- 0 → unicast
- 1 → multicast

Example:

- 00:11:22:33:44:55 → unicast.
- 01:00:5E:xx:xx:xx → multicast

LSB of first octet determines unicast or multicast.

ii) The statement is incorrect.

MAC Address are:

→ flat address.

→ not hierarchical

→ hardware - based.

They are portable only if the device moves with its NIC.

IP Address are hierarchical